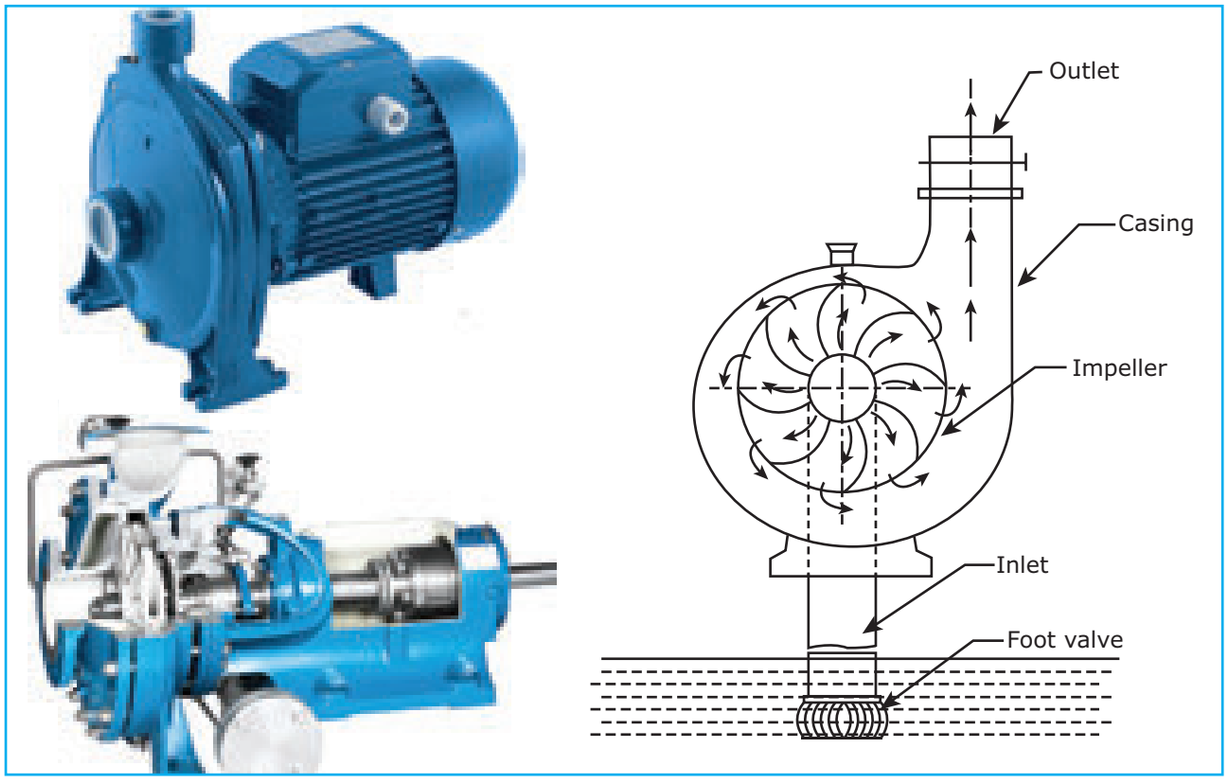


Centrifugal pump

Working principles of centrifugal pump

Before the pump is set on for pumping, priming is done. Priming is done by filling water is the suction pipe and the impeller. While doing so, the air valve is opened. As a result, no air pocket is left in the casing. The presence of any air will certainly decrease the sucking capacity of the impeller.

After the priming is over, the pump is started. The prime mover operates the impeller. The vane on the impeller create centrifugal force inside the casing. Vacuum is created at the centre of the impeller (eye). Water from the reservoir reaches the casing through the inlet. Due to the centrifugal force, water is thrown radically outwards. As the sectional area of the casing gradually increases, velocity energy of the liquid is transformed into pressure energy. The liquid with increases pressure is pumped to a required height through the outlet. As the impeller goes on rotating, more water is pumped out.

Application of centrifugal pump

- Water pump houses
- Sewage treatment
- Irrigation
- Oil refineries

Advantages

- Durability
- Low cost
- Suitable for pumping coolant

Disadvantages

- Low efficiency
- Limited to be used at low pressure

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